

# Jayden Melendez

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## Education

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**Marist College**, Poughkeepsie, New York

B.S. in Computer Science

Minor: Cyber Security

Graduation: May 2026

GPA: 3.95

## Relevant Coursework

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Operating Systems, Algorithms, Computer Organization and Architecture, Intro to z/OS & Major Subsystems, Internetworking, System Design, Software Development II, Database Management, Mobile Security, Discrete Mathematics

## Professional Experience

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**International Business Machine (IBM)**, Poughkeepsie, New York

*Backend Developer Co-op*

May 2024 – Present

- **Architectural Modernization:** Contributed to the refactoring of a legacy **50,000+ LoC Java** mainframe patch management system; transitioned tightly-coupled modules into a **Service-Oriented Architecture (SOA)** to improve horizontal scalability and maintainability.
- **Systems Engineering:** Engineered a programmatic diagnostic engine that parses data across hundreds of relational database tables to automate hardware issue detection; reduced manual troubleshooting time by **10%** for 50+ global support.
- **Performance Optimization:** Optimized complex **SQL queries and indexing strategies** within a DB2 environment, achieving a **20% reduction** in existing data retrieval latency for critical patch lifecycle tracking.
- **Reliability & DevSecOps:** Enhanced system stability by implementing a comprehensive **SwaggerUI** testing suite and utilized **Podman** for containerized development, ensuring codebase environmental consistency among team members.
- **Collaborative Leadership:** Actively collaborate with senior architects to identify technical debt, utilizing **Agile methodologies** and **Github Enterprise** for code reviews and version control to ensure high-quality software delivery.

**Marist College**, Poughkeepsie, New York

*Resident Assistant*

Aug 2023 – Present

- Manage crisis intervention, conflict resolution, and facility logistics for a community of 60+ residents.
- Facilitate community engagement and budget management through coordinated educational programming.

## Projects

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**6502 Operating System Emulator**

Aug 2025 - Dec 2025

*Typescript, Node.js*

- Studied and referenced the fundamentals of operating system architecture to build a fully functioning OS emulator.
- Features include a user interface to complete computational actions on behalf of the user.

**NutritionU - A meal planner for students**

Aug 2025 – Dec 2025

*React, Vite, Node.js, Postgres, Postman*

- Utilized modern web development skills to actively build a full stack web application that gives students a road map for cost effective and healthy eating.
- Used the Open AI application programming interface to build features such as modifying recipes for substitutions in the case of dietary restrictions, calculating the estimated number of macro-nutrients a user should be eating, and utilizing the ingredients in a user's grocery list to build a meal plan that fits within the applications personalized recommendations.
- Designed a relational database from scratch using PostgreSQL to query user data. This data is then analyzed and processed securely through robust data modeling practices.

## Skills

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**Programming Languages:** JavaScript, SQL, Python, Java, TypeScript, C++, Assembly, HTML/CSS

**Technologies/Frameworks:** React, ReactNative, Springboot, ExpoGo, Node.js, Podman, IBM z/OS, Linux, Git, JSP

**Developer Tools:** Visual Studio Code, Postman API, GitHub Enterprise, SwaggerUI

## Extracurriculars

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**Ultimate Frisbee Player**, Marist College, Poughkeepsie, NY

Aug 2022 – Present

**Spiritual Life and Service Volunteer**, Marist College, Poughkeepsie, NY

Aug 2022 – Present